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Exploring the grey zone of E/e' ratio: does left atrial strain help?

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Background: The combination of early trans-mitral inflow and mitral annular tissue Doppler velocities (E/e' ratio) is widely applied to noninvasively estimate left ventricular (LV) filling pressures. However E/e' ratio has a significant gray zone that restricts its accuracy and left atrial (LA) deformation analysis by speckle tracking echocardiography (STE) was recently proposed as an alternative approach to estimate LV filling pressures, but the clinical application of LA strain in the subgroup of patients with E/e' between 8 and 14 has been under-investigated.

Aims: This study aimed to analyze the role of LA longitudinal function by STE (PALS) to estimate intra-cardiac pressures as assessed by systolic pulmonary artery pressure (sPAP), measured by Doppler, specifically in patients with an E/e' ratio >8 and ≤ 14 .

Methods: We enrolled 142 consecutive, non-selected patients, referred to our echocardiography laboratory for a comprehensive transthoracic echocardiography. Exclusion criteria were: organic mitral valve disease or prosthesis and presence of disease possibly associated with pre-capillary pulmonary hypertension. Particular care was used for accurate measurement

of maximal tricuspid regurgitation velocity and of right atrial pressure and consequently sPAP estimation. PALS values were obtained by averaging all segments, and by separately averaging segments measured in the 4-chamber and 2-chamber views.

Results: Seventy-four patients (52% of total) showed an E/e' ratio >8 and ≤ 14 , with the following characteristics: mean age 65.5 ± 11.9 years, LVEF 54.5 ± 11.2 , E/e' 11.2 ± 1.9 , sPAP 33 ± 7 mmHg, PALS $31.6 \pm 11.7\%$. A negative correlation between PALS and sPAP was found ($r = -0.55$, $p < 0.0001$). From receiver operating characteristic (ROC) curves, PALS demonstrated a high diagnostic accuracy (AUC 0.78 (95% CI: 66%–90%)); the cutoff value of 23% showed an excellent specificity of 90% with a sensibility of 60%, to predict sPAP higher than 35 mmHg.

Conclusions: LA function measured by STE is a simple parameter able to predict increased intra-cardiac pressure even in the intermediate E/e' group. This parameters might help in improving the diagnostic algorithm of diastolic function.

